Week 5 – MATH I

Fall 2026

Continuity (function continuity)

Section 2.4 from textbook

&

Continuity, Inverse Functions, and Weierstrass' Theorem¶

Instructor's NOTES (on Moodle)

Lecture presentation prepared and presented by G. Neofotistos

...from textbook

2.4 Continuity

Learning Objectives

2.4.1 Explain the three conditions for continuity at a point.
2.4.2 Describe three kinds of discontinuities.
2.4.3 Define continuity on an interval.
2.4.4 State the theorem for limits of composite functions.
2.4.5 Provide an example of the intermediate value theorem.

Many functions have the property that their graphs can be traced with a pencil without lifting the pencil from the page. Such functions are called *continuous*. Other functions have points at which a break in the graph occurs, but satisfy this property over intervals contained in their domains. They are continuous on these intervals and are said to have a *discontinuity at a point* where a break occurs.

3 conditions for the continuity of a function at a point

A function f is continuous at $x = a$ if

$$\boxed{\lim_{x \to a} f(x) = f(a)}.$$

This condition contains three requirements:

1. $f(a)$ is defined.
2. $\lim_{x \to a} f(x)$ exists.
3. The limit equals the function value.

Equivalently,

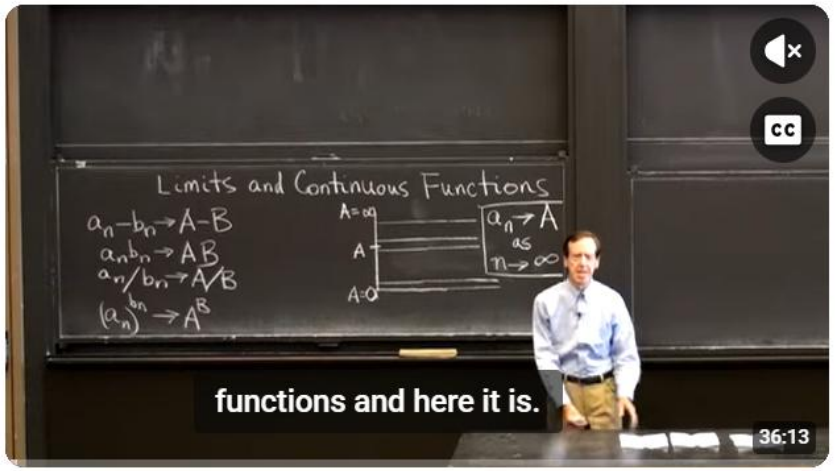
$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a).$$

Extra (multimedia) material

function continuity

Create

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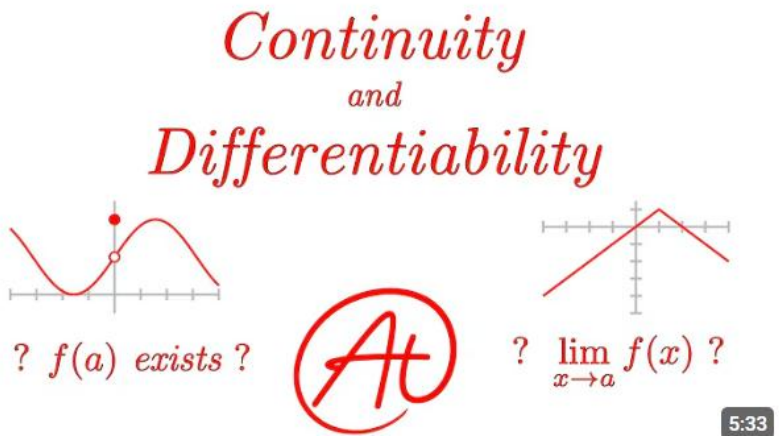
Limits and Continuous Functions

287K • 16y ago

MIT OpenCourseWare

Limits and **Continuous Functions** Instructor: Gilbert Strang http://ocw.mit.edu/highlights-of-calculus License: Creative Commo

CC



Continuity and Differentiability EXPLAINED with Examples

288K • 4y ago

Ace Tutors

Learn the meaning behind **Continuity** and Differentiability. This video explains what **Continuity** is and the 3 rules for a **function**

4K **College**

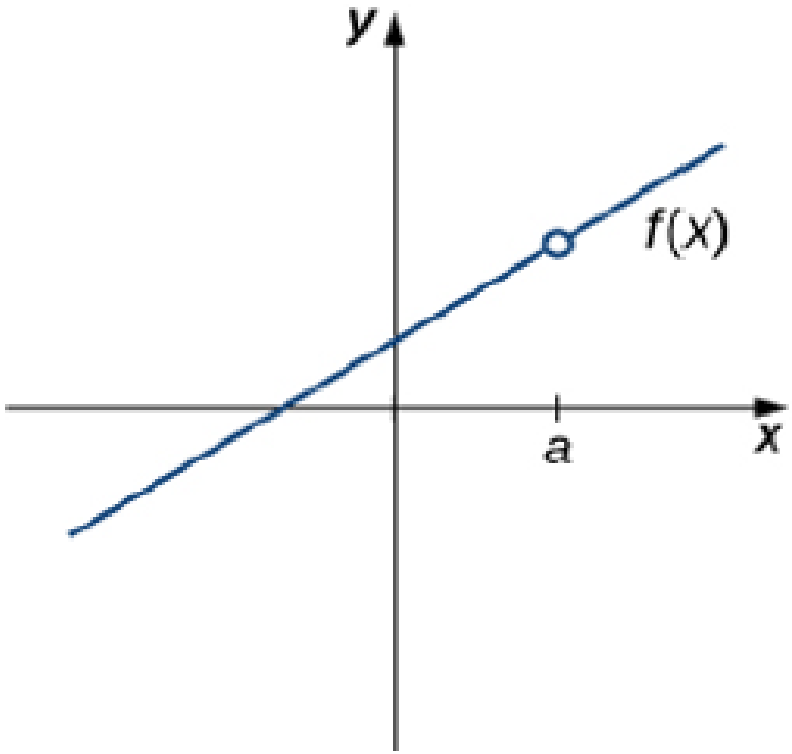
Let’s play ½ of the second YouTube presentation

https://www.youtube.com/watch?v=nRv1lPqrx3k

In-Class activity

Is the function (shown below) continuous?

Explain



In-Class activity

Is the function (shown below) continuous?

Explain

i. $f(a)$ is defined.

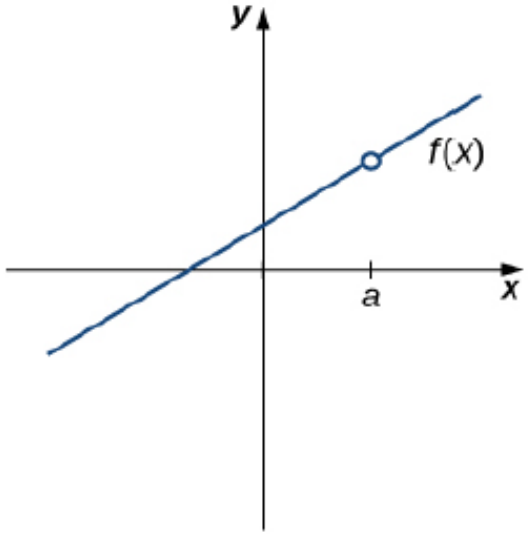
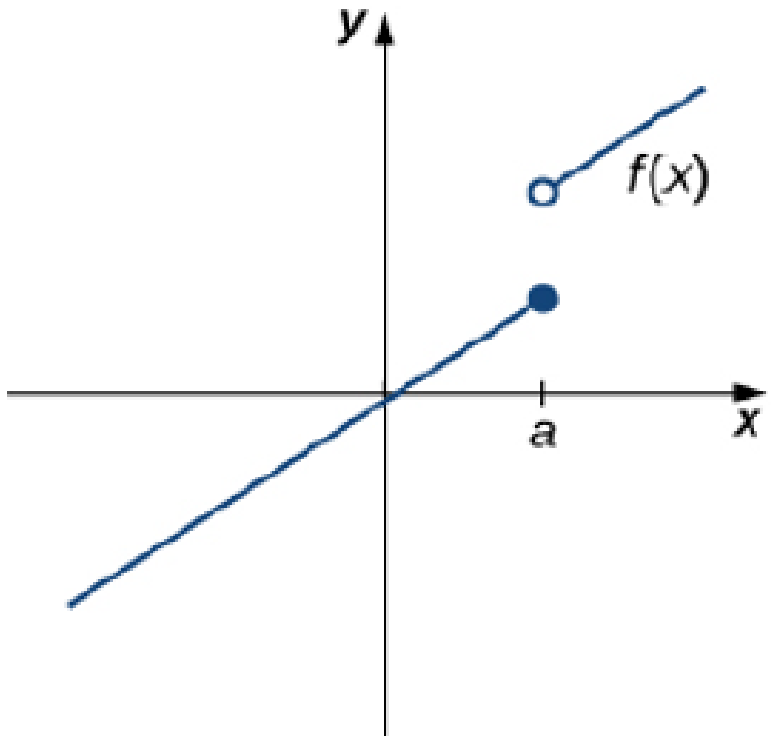


Figure 2.32 The function $f(x)$ is not continuous at a because $f(a)$ is undefined.

In-Class activity

Is the function (shown below) continuous?

Explain



In-Class activity

Is the function (shown below) continuous?

Explain

ii. $\lim_{x \to a} f(x)$ exists.

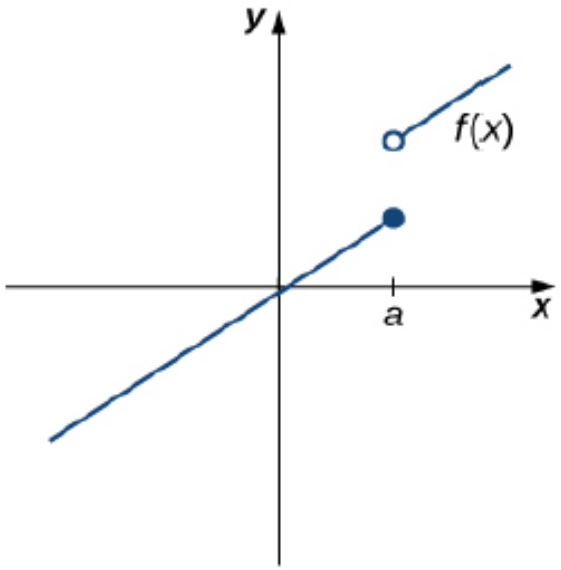
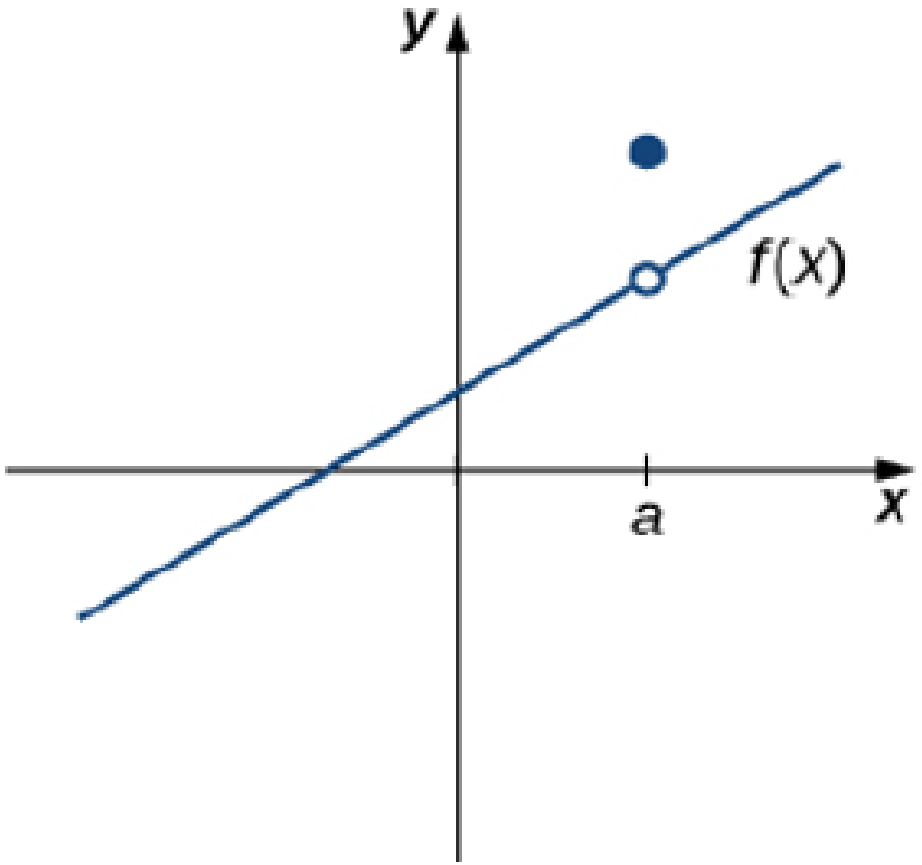


Figure 2.33 The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x)$ does not exist.

In-Class activity

Is the function (shown below) continuous?

Explain



In-Class activity

Is the function (shown below) continuous?

Explain

iii. $\lim_{x \to a} f(x) = f(a)$.

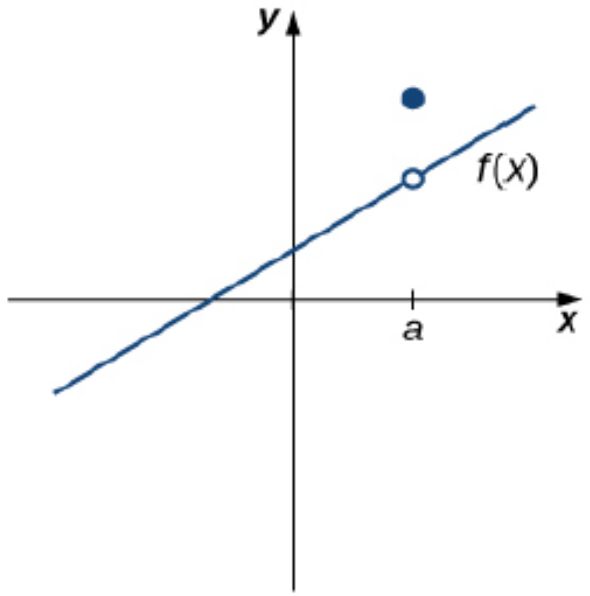


Figure 2.34 The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x) \neq f(a)$.

Definition

A function $f(x)$ is **continuous at a point** *a* if and only if the following three conditions are satisfied:

i. $f(a)$ is defined
ii. $\lim_{x \to a} f(x)$ exists
iii. $\lim_{x \to a} f(x) = f(a)$

A function is **discontinuous at a point** *a* if it fails to be continuous at *a*.

The following procedure can be used to analyze the continuity of a function at a point using this definition.

Problem-Solving Strategy

Determining Continuity at a Point

1. Check to see if $f(a)$ is defined. If $f(a)$ is undefined, we need go no further. The function is not continuous at *a*. If $f(a)$ is defined, continue to step 2.
2. Compute $\lim_{x \to a} f(x)$. In some cases, we may need to do this by first computing $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. If $\lim_{x \to a} f(x)$ does not exist (that is, it is not a real number), then the function is not continuous at *a* and the problem is solved. If $\lim_{x \to a} f(x)$ exists, then continue to step 3.
3. Compare $f(a)$ and $\lim_{x \to a} f(x)$. If $\lim_{x \to a} f(x) \neq f(a)$, then the function is not continuous at *a*. If $\lim_{x \to a} f(x) = f(a)$, then the function is continuous at *a*.

In-Class activity

Justify your conclusion

Determining Continuity at a Point, Condition 1

Using the definition, determine whether the function $f(x) = (x^2 - 4)/(x - 2)$ is continuous at $x = 2$. Justify the conclusion.

Solution

Determining Continuity at a Point, Condition 1

Using the definition, determine whether the function $f(x) = (x^2 - 4)/(x - 2)$ is continuous at $x = 2$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(2)$. We can see that $f(2) = 0/0$, which is undefined. Therefore, $f(x) = \frac{x^2-4}{x-2}$ is discontinuous at 2 because $f(2)$ is undefined. The graph of $f(x)$ is shown in Figure 2.35.

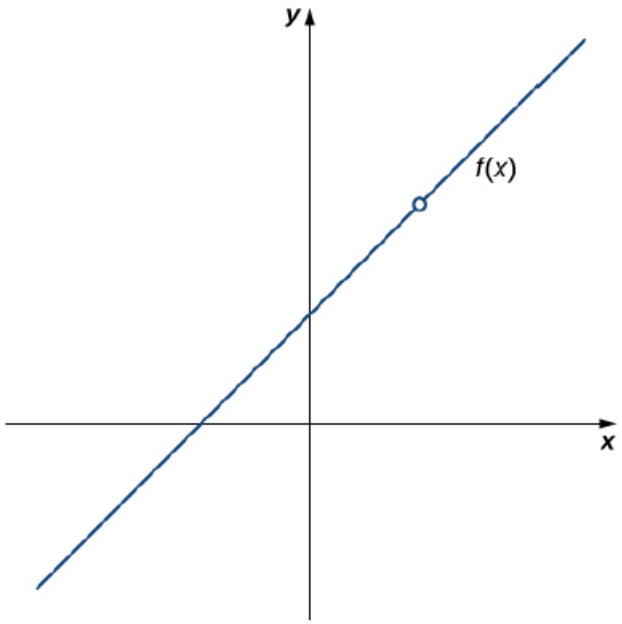


Figure 2.35 The function $f(x)$ is discontinuous at 2 because $f(2)$ is undefined.

In-Class activity

Justify your conclusion

Determining Continuity at a Point, Condition 2

Using the definition, determine whether the function $f(x) = \begin{cases} -x^2 + 4 & \text{if } x \leq 3 \\ 4x - 8 & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. Justify the conclusion.

Solution

Determining Continuity at a Point, Condition 2

Using the definition, determine whether the function $f(x) = \begin{cases} -x^2 + 4 & \text{if } x \leq 3 \\ 4x - 8 & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(3)$.

$$f(3) = -(3^2) + 4 = -5.$$

Thus, $f(3)$ is defined. Next, we calculate $\lim_{x \to 3} f(x)$. To do this, we must compute $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3^+} f(x)$:

$$\lim_{x \to 3^-} f(x) = -(3^2) + 4 = -5$$

and

$$\lim_{x \to 3^+} f(x) = 4(3) - 8 = 4.$$

Therefore, $\lim_{x \to 3} f(x)$ does not exist. Thus, $f(x)$ is not continuous at 3. The graph of $f(x)$ is shown in Figure 2.36.

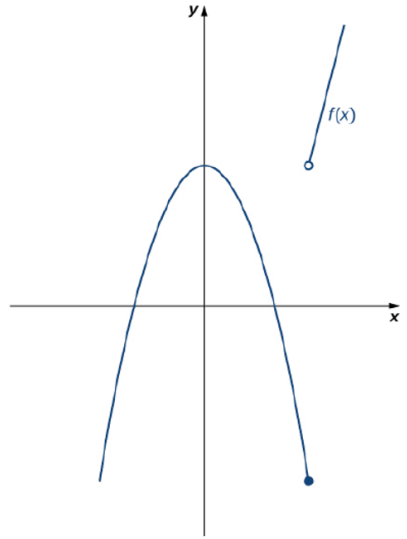


Figure 2.36 The function $f(x)$ is not continuous at 3 because $\lim_{x \to 3} f(x)$ does not exist.

In-Class activity
Justify your conclusion

Determining Continuity at a Point, Condition 3

Using the definition, determine whether the function $f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.

In-Class activity Solution

Determining Continuity at a Point, Condition 3

Using the definition, determine whether the function $f(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.

Solution

First, observe that

$$f(0) = 1.$$

Next,

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin x}{x} = 1.$$

Last, compare $f(0)$ and $\lim_{x \to 0} f(x)$. We see that

$$f(0) = 1 = \lim_{x \to 0} f(x).$$

Since all three of the conditions in the definition of continuity are satisfied, $f(x)$ is continuous at $x = 0$.

In-Class activity

Find the value of k that makes

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3, \\ k, & x = 3 \end{cases}$$

continuous at $x = 3$.

Find the value of k that makes

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3, \\ k, & x = 3 \end{cases}$$

continuous at $x = 3$.

Solution

For $x \neq 3$,

$$f(x) = \frac{(x-3)(x+3)}{x-3} = x + 3.$$

Therefore,

$$\lim_{x \to 3} f(x) = 6.$$

Continuity requires

$$k = \lim_{x \to 3} f(x).$$

Hence,

$$\boxed{k = 6}.$$

Solution

Theorem 2.8

Continuity of Polynomials and Rational Functions

Polynomials and rational functions are continuous at every point in their domains.

This is very helpful for Engineering!

EXAMPLE 2.29

Continuity of a Rational Function

For what values of x is $f(x) = \frac{x+1}{x-5}$ continuous?

Solution

The rational function $f(x) = \frac{x+1}{x-5}$ is continuous for every value of x except $x = 5$.

Theorem 2.10

Continuity of Trigonometric Functions

Trigonometric functions are continuous over their entire domains.

Algebra of continuous functions

Very useful for Engineering!

If f and g are continuous at $x = a$, then the following functions are also continuous at a:

$f + g, \qquad f - g, \qquad cf, \qquad fg,$

and

$\frac{f}{g},$

provided that $g(a) \neq 0$.

Important to know: Polynomial functions are continuous everywhere. Rational functions are continuous wherever their denominators are nonzero. Exponential, logarithmic and trigonometric functions are continuous on their domains.

1. Continuity ¶

1.1 Informal meaning

A function is continuous if its graph has no breaks, jumps, holes, or vertical asymptotes within the region being considered.

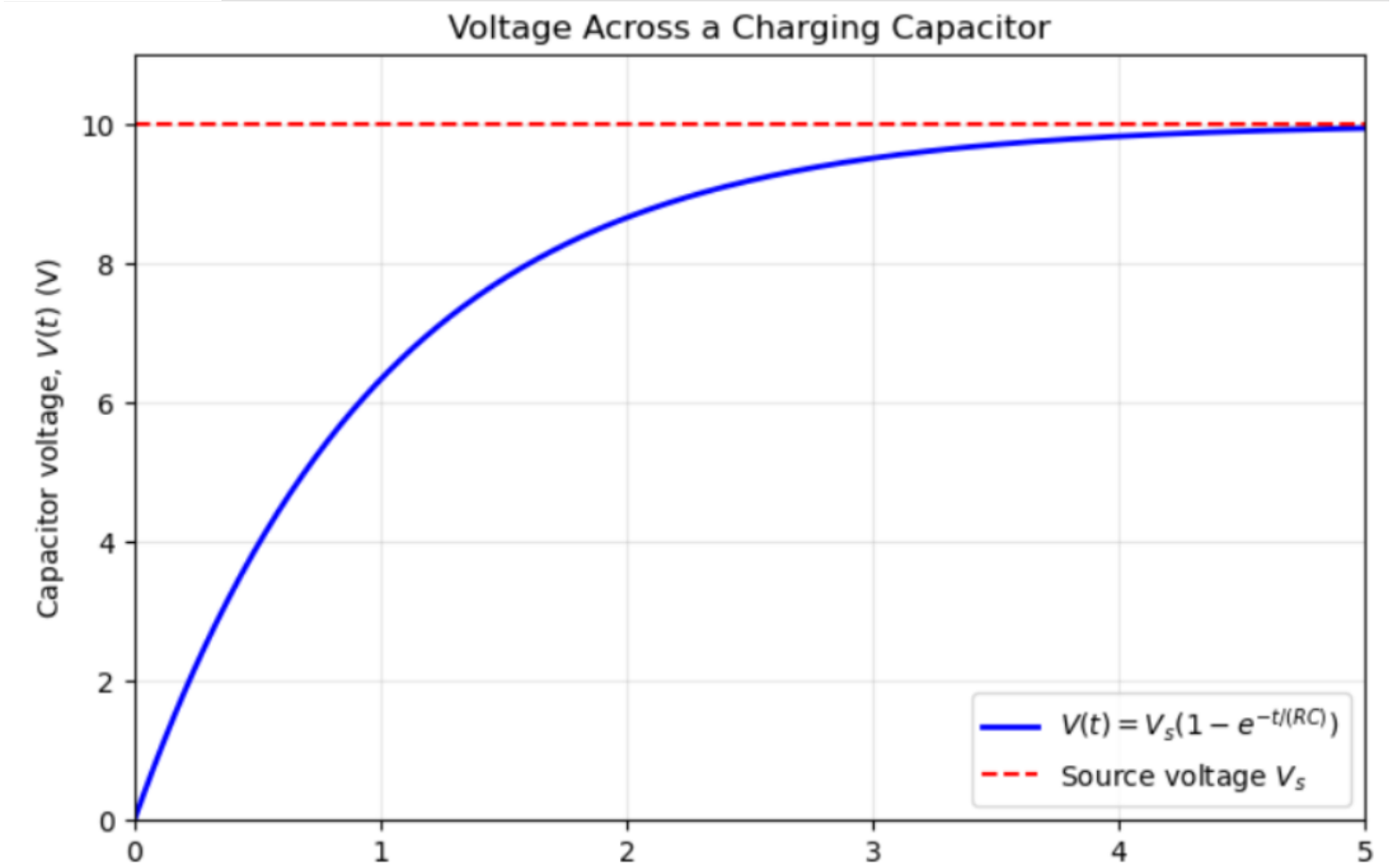
In engineering terms, if $y = f(x)$ is continuous near $x = a$, then a sufficiently small change in the input x produces a small change in the output y.

For example, the voltage across a charging capacitor,

$$V(t) = V_s\left(1 - e^{-t/(RC)}\right),$$

is continuous for $t \geq 0$. The model predicts that the voltage changes smoothly rather than jumping instantaneously (see plot below).

...from NOTES



Types of Discontinuities

As we have seen in Example 2.26 and Example 2.27, discontinuities take on several different appearances. We classify the types of discontinuities we have seen thus far as removable discontinuities, infinite discontinuities, or jump discontinuities. Intuitively, a **removable discontinuity** is a discontinuity for which there is a hole in the graph, a **jump discontinuity** is a noninfinite discontinuity for which the sections of the function do not meet up, and an **infinite discontinuity** is a discontinuity located at a vertical asymptote. Figure 2.37 illustrates the differences in these types of discontinuities. Although these terms provide a handy way of describing three common types of discontinuities, keep in mind that not all discontinuities fit neatly into these categories.

Important classification of discontinuities

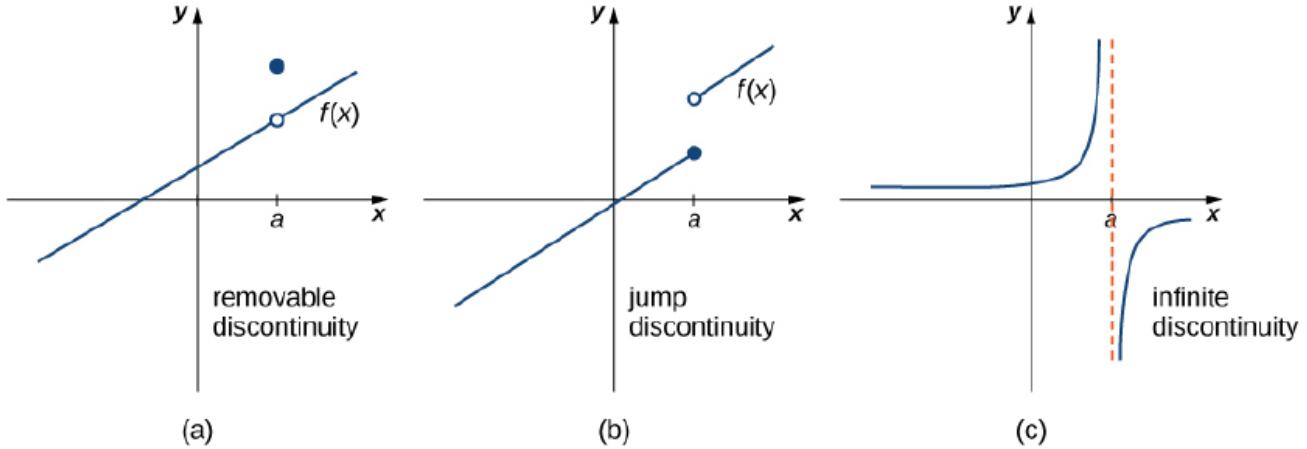


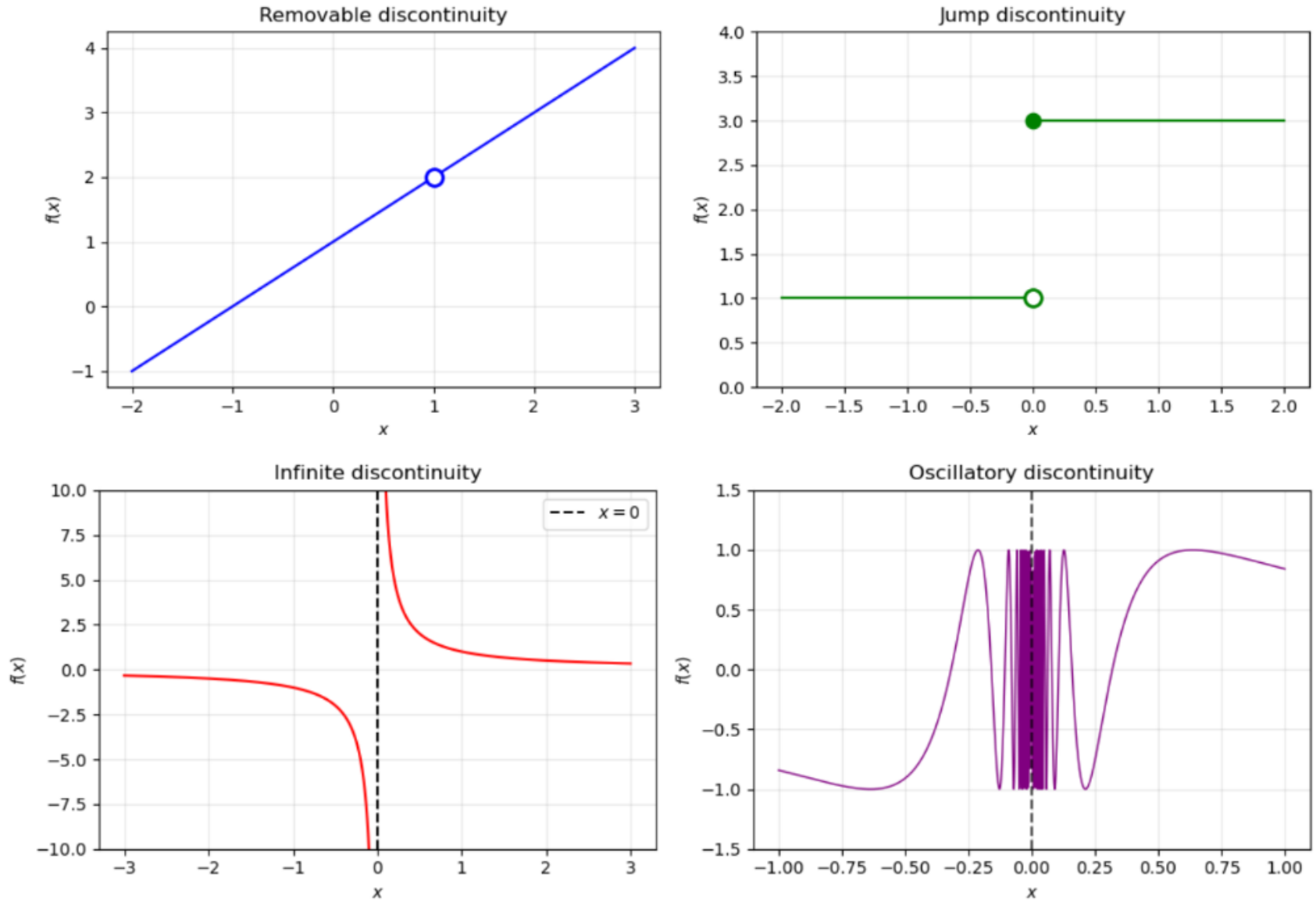
Figure 2.37 Discontinuities are classified as (a) removable, (b) jump, or (c) infinite.

These three discontinuities are formally defined as follows:

Definition

If $f(x)$ is discontinuous at a, then

1. f has a **removable discontinuity** at a if $\lim_{x \to a} f(x)$ exists. (Note: When we state that $\lim_{x \to a} f(x)$ exists, we mean that $\lim_{x \to a} f(x) = L$, where L is a real number.)
2. f has a **jump discontinuity** at a if $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist, but $\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$. (Note: When we state that $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ both exist, we mean that both are real-valued and that neither take on the values $\pm\infty$.)
3. f has an **infinite discontinuity** at a if $\lim_{x \to a^-} f(x) = \pm\infty$ and/or $\lim_{x \to a^+} f(x) = \pm\infty$.


Main Types of Function Discontinuities
Removable discontinuity
Jump discontinuity
Infinite discontinuity
Oscillatory discontinuity
f(x)
x
x = 0

Continuity on an Interval

State the interval(s) over which the function $f(x) = \frac{x-1}{x^2+2x}$ is continuous.

Solution

Since $f(x) = \frac{x-1}{x^2+2x}$ is a rational function, it is continuous at every point in its domain. The domain of $f(x)$ is the set $(-\infty, -2) \cup (-2, 0) \cup (0, +\infty)$. Thus, $f(x)$ is continuous over each of the intervals $(-\infty, -2)$, $(-2, 0)$, and $(0, +\infty)$.

Continuity over an Interval

State the interval(s) over which the function $f(x) = \sqrt{4 - x^2}$ is continuous.

Solution

From the limit laws, we know that $\lim_{x \to a} \sqrt{4 - x^2} = \sqrt{4 - a^2}$ for all values of a in $(-2, 2)$. We also know that $\lim_{x \to -2^+} \sqrt{4 - x^2} = 0$ exists and $\lim_{x \to 2^-} \sqrt{4 - x^2} = 0$ exists. Therefore, $f(x)$ is continuous over the interval $[-2, 2]$.

About Composite functions

Theorem 2.9

Composite Function Theorem

If $f(x)$ is continuous at L and $\lim_{x \to a} g(x) = L$, then

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right) = f(L).$$

Before we move on to Example 2.35, recall that earlier, in the section on limit laws, we showed $\lim_{x \to 0} \cos x = 1 = \cos(0)$. Consequently, we know that $f(x) = \cos x$ is continuous at 0. In Example 2.35 we see how to combine this result with the composite function theorem.

EXAMPLE 2.35

Limit of a Composite Cosine Function

Evaluate $\lim_{x \to \pi/2} \cos\left(x - \frac{\pi}{2}\right)$.

Solution

The given function is a composite of $\cos x$ and $x - \frac{\pi}{2}$. Since $\lim_{x \to \pi/2} \left(x - \frac{\pi}{2}\right) = 0$ and $\cos x$ is continuous at 0, we may apply the composite function theorem. Thus,

$$\lim_{x \to \pi/2} \cos\left(x - \frac{\pi}{2}\right) = \cos\left(\lim_{x \to \pi/2} \left(x - \frac{\pi}{2}\right)\right) = \cos(0) = 1.$$

The Intermediate Value Theorem

Functions that are continuous over intervals of the form $[a, b]$, where a and b are real numbers, exhibit many useful properties. Throughout our study of calculus, we will encounter many powerful theorems concerning such functions. The first of these theorems is the **Intermediate Value Theorem**.

Important for Engineering!

Theorem 2.11

The Intermediate Value Theorem

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$ in Figure 2.38.

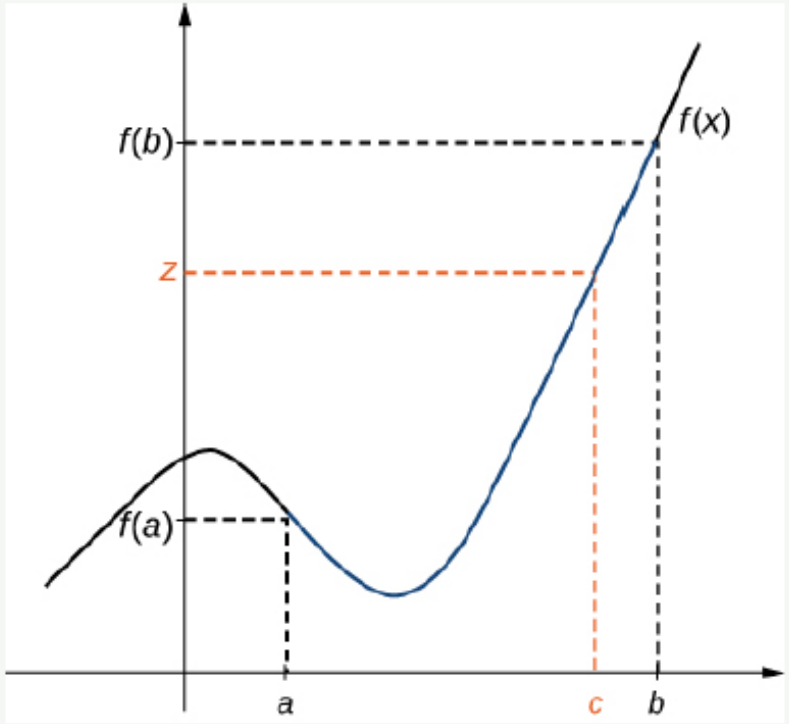


Figure 2.38 There is a number $c \in [a, b]$ that satisfies $f(c) = z$.

Example (from NOTES)
Engineering meaning

5. Intermediate Value Theorem

The Intermediate Value Theorem provides an important consequence of continuity.

> If f is continuous on $[a, b]$, and L lies between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that
>
> $$f(c) = L.$$

In particular, if

$f(a)f(b) < 0,$

then $f(a)$ and $f(b)$ have opposite signs, so there exists at least one $c \in (a, b)$ for which

$f(c) = 0.$

The theorem guarantees existence, but not uniqueness.

Engineering meaning

If the temperature of a *continuously operating system* rises from $40^\circ\mathrm{C}$ to $80^\circ\mathrm{C}$, then it must pass through every intermediate temperature, including $60^\circ\mathrm{C}$.

This conclusion would not necessarily hold for a discontinuous model.

EXAMPLE 2.36

Application of the Intermediate Value Theorem

Show that $f(x) = x - \cos x$ has at least one zero.

Solution

Since $f(x) = x - \cos x$ is continuous over $\left(-\infty, +\infty\right)$, it is continuous over any closed interval of the form $[a, b]$. If you can find an interval $[a, b]$ such that $f(a)$ and $f(b)$ have opposite signs, you can use the Intermediate Value Theorem to conclude there must be a real number c in (a, b) that satisfies $f(c) = 0$. Note that

$$f(0) = 0 - \cos(0) = -1 < 0$$

and

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} - \cos\frac{\pi}{2} = \frac{\pi}{2} > 0.$$

Using the Intermediate Value Theorem, we can see that there must be a real number c in $[0, \pi/2]$ that satisfies $f(c) = 0$. Therefore, $f(x) = x - \cos x$ has at least one zero.

In-Class activity

When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

Solution

When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

Solution

No. The Intermediate Value Theorem only allows us to conclude that we can find a value between $f(0)$ and $f(2)$; it doesn't allow us to conclude that we can't find other values. To see this more clearly, consider the function $f(x) = (x-1)^2$. It satisfies $f(0) = 1 > 0$, $f(2) = 1 > 0$, and $f(1) = 0$.

In-Class activity

EXAMPLE 2.37

When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

Solution

EXAMPLE 2.37

When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

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No. The Intermediate Value Theorem only allows us to conclude that we can find a value between $f(0)$ and $f(2)$; it doesn't allow us to conclude that we can't find other values. To see this more clearly, consider the function $f(x) = (x - 1)^2$. It satisfies $f(0) = 1 > 0$, $f(2) = 1 > 0$, and $f(1) = 0$.

In-Class activity

EXAMPLE 2.38

When Can You Apply the Intermediate Value Theorem?

For $f(x) = 1/x$, $f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Can we conclude that $f(x)$ has a zero in the interval $[-1, 1]$?

Solution

EXAMPLE 2.38

When Can You Apply the Intermediate Value Theorem?

For $f(x) = 1/x$, $f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Can we conclude that $f(x)$ has a zero in the interval $[-1, 1]$?

Solution

No. The function is not continuous over $[-1, 1]$. The Intermediate Value Theorem does not apply here.

What to remember about the Intermediate Value Theorem (IVT)

The IVT depends only on continuity on an interval:

> If f is continuous on $[a, b]$, then it takes every value between $f(a)$ and $f(b)$.

Continuity and Inverse functions

4. Inverse functions

4.1 Meaning of an inverse

Suppose

$y = f(x)$.

An inverse function reverses the input-output relationship:

$x = f^{-1}(y)$.

The defining properties are

$f^{-1}(f(x)) = x$

and

$f(f^{-1}(y)) = y$.

The notation $f^{-1}(x)$ does **not** usually mean $1/f(x)$:

$$\boxed{f^{-1}(x) \neq \frac{1}{f(x)}}.$$

For example, if

$f(x) = 3x + 2$,

then solving $y = 3x + 2$ for x gives

$x = \frac{y-2}{3}$.

Hence,

$f^{-1}(y) = \frac{y-2}{3}$.

4.2 When does an inverse function exist?

A function has an inverse on its domain if it is **one-to-one**:

$f(x_1) = f(x_2) \implies x_1 = x_2$.

Graphically, a one-to-one function passes the horizontal-line test: every horizontal line intersects its graph at most once.

A continuous function that is strictly increasing or strictly decreasing on an interval is one-to-one there and therefore has an inverse.

For example, $f(x) = x^2$ is not one-to-one on $\mathbb{R}$, because

$f(-2) = f(2) = 4$.

However, if its domain is restricted to $[0, \infty)$, then it is strictly increasing and

$f^{-1}(x) = \sqrt{x}$.

Monotonic functions

A function is **monotonic** on an interval if its values consistently move in one direction as x increases.

A function f is:

- **increasing** if

$$x_1 < x_2 \implies f(x_1) \leq f(x_2);$$

- **strictly increasing** if

$$x_1 < x_2 \implies f(x_1) < f(x_2);$$

- **decreasing** if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2);$$

- **strictly decreasing** if

$$x_1 < x_2 \implies f(x_1) > f(x_2).$$

A function that is strictly increasing or strictly decreasing is called **strictly monotonic**.

Strict monotonicity implies one-to-one

Every strictly monotonic function is one-to-one. For example, if f is strictly increasing and $x_1 < x_2$, then

$$f(x_1) < f(x_2).$$

Therefore, two different inputs cannot produce the same output.

Continuity of Inverse functions & an example

The principal theorem is:

> If f is continuous and strictly monotonic on an interval I, then f is one-to-one, and its inverse
>
> $$f^{-1} : f(I) \to I$$
>
> exists and is continuous.

In short,

$$\boxed{\text{continuous} + \text{strictly monotonic} \implies \text{continuous inverse.}}$$

Continuity ensures that f does not skip intermediate output values, while strict monotonicity ensures that it does not repeat them:

$$\underbrace{\text{continuity}}_{\text{no gaps}} + \underbrace{\text{strict monotonicity}}_{\text{no repeated values}} \implies \underbrace{\text{continuous inverse}}_{\text{stable reversal}}.$$

The function

$$f(x) = e^x$$

is continuous and strictly increasing on $\mathbb{R}$. Its range is $(0, \infty)$. Therefore, it has a continuous inverse:

$$\boxed{f^{-1}(x) = \ln x, \qquad x > 0.}$$

The points on the two graphs are reversed. For example,

$$e^0 = 1$$

means that $(0, 1)$ lies on $y = e^x$, while

$$\ln 1 = 0$$

means that $(1, 0)$ lies on $y = \ln x$.

Example (from NOTES)

4.4 Engineering interpretation

Suppose a sensor output is related to temperature by

$V = f(T)$.

If f is continuous and strictly increasing, then:

- each temperature produces exactly one voltage;
- each voltage in the sensor's range corresponds to exactly one temperature;
- the calibration can be inverted:

$T = f^{-1}(V)$;

- small changes in the measured voltage produce small changes in the recovered temperature.

Continuity alone is not sufficient for invertibility. The calibration curve must also be one-to-one.

Example: Inverting a sensor model

Another example

Suppose a sensor follows

$$V(T) = 2 + 0.01T^2, \qquad T \geq 0.$$

This function is *continuous and strictly increasing* for $T \geq 0$. *Therefore, it has a continuous inverse.*

Solve for T:

$$V - 2 = 0.01T^2,$$

so

$$T^2 = 100(V - 2).$$

Because $T \geq 0$,

$$\boxed{T = V^{-1}(V) = 10\sqrt{V - 2}}, \qquad V \geq 2.$$

Here, it is clearer to name the inverse calibration function $T(V)$:

$$\boxed{T(V) = 10\sqrt{V - 2}}.$$

For $V = 11\ \mathrm{V}$,

$$T = 10\sqrt{11 - 2} = 30.$$

Thus, the measured temperature is $30^\circ\mathrm{C}$.

...one more example

The displacement x of an actuator is related to input voltage V by

$$x(V) = 0.5V + 2, \qquad 0 \le V \le 10.$$

1. Find the range of x.
2. Find the inverse $V(x)$.
3. Explain why the inverse is continuous.

Solution

At the endpoints,

$$x(0) = 2, \qquad x(10) = 7.$$

Because $x(V)$ is increasing, its range is

$$[2, 7].$$

To find the inverse, solve

$$x = 0.5V + 2.$$

Then

$$x - 2 = 0.5V,$$

so

$$\boxed{V(x) = 2(x - 2) = 2x - 4}, \qquad 2 \le x \le 7.$$

The original function is continuous and strictly increasing on $[0, 10]$. Therefore, its inverse is continuous on $[2, 7]$.

https://en.wikipedia.org/wiki/Karl_Weierstrass

Karl Weierstrass

71 languages

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From Wikipedia, the free encyclopedia

Karl Theodor Wilhelm Weierstrass (/ˈvaɪər ˌstrɑːs, -ˌʃtrɑːs/;[1] German: *Weierstraß* [ˈvaɪɐʃtʁaːs];[2] 31 October 1815 – 19 February 1897) was a German mathematician often cited as the "father of modern analysis".[3] Despite leaving university without a degree, he studied mathematics and trained as a school teacher, eventually teaching mathematics, physics, botany and gymnastics.[4] He later received an honorary doctorate and became professor of mathematics in Berlin.

Among many other contributions, Weierstrass formalized the definition of the continuity of a function and complex analysis, proved the intermediate value theorem and the Bolzano–Weierstrass theorem, and used the latter to study the properties of continuous functions on closed bounded intervals.

Karl Weierstrass
Karl Weierstraß



Is the Bolzano-Weierstrass theorem the same as the Weierstrass theorem?

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AI Overview

No, the Bolzano-Weierstrass theorem is not the same as the Extreme Value Theorem (often called the Weierstrass extreme value theorem), though both are foundational results in mathematical analysis named in part after Karl Weierstrass. Wikipedia +1

6. Weierstrass' Extreme Value Theorem

In introductory calculus, "Weierstrass' Theorem" usually refers to the **Extreme Value Theorem**.

> If f is continuous on a closed and bounded interval $[a, b]$, then f attains both an absolute maximum and an absolute minimum on that interval.

Thus, there exist points $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for every $x \in [a, b]$.

Notice the word **attains**. The theorem says that the maximum and minimum are actual function values, not merely values that the function approaches.

Why the hypotheses matter

Both conditions are important:

1. The function must be continuous.
2. The interval must be closed and bounded.

Finding absolute extrema in practice

For a function that is continuous on $[a, b]$:

1. Find the critical points inside (a, b), usually by solving

$f'(x) = 0$

(wait till next week, when we are going to introduce derivatives!)

2. Evaluate f at every critical point.
3. Evaluate $f(a)$ and $f(b)$.
4. Compare all values.

The largest value is the absolute maximum, and the smallest is the absolute minimum.

Weierstrass' theorem guarantees that these extrema exist. Differential calculus helps us locate them.

How important the Weierstrass theorem can be for engineering students?

Weierstrass' Extreme Value Theorem is very important in engineering—not mainly as a computational technique, but as an **existence guarantee**.

If an engineering performance function f is continuous and the allowable design set is closed and bounded, then an optimal design actually exists:

$$\boxed{\textbf{continuous model} + \textbf{closed bounded feasible set} \Longrightarrow \textbf{attained maximum and minimum.}}$$

This principle underlies structural optimization, energy management, control, circuit design, thermal analysis, materials selection and numerical optimization. Before trying to calculate an optimum, an engineer should ask whether an optimum is guaranteed to exist.

For a continuous function $f : [a, b] \to \mathbb{R}$, Weierstrass' theorem states that there are $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for every $x \in [a, b]$.

The theorem guarantees **existence**, but does not tell us:

- where the extrema occur;
- whether they are unique;
- how to calculate them;
- whether a local extremum is also global.

Derivatives, numerical methods and comparisons of candidate values are needed for those tasks.

A deeper engineering interpretation

Weierstrass' theorem supports three important engineering statements:

1. **An optimal design exists.**
   If the objective is continuous and the feasible design set is compact, the best design is not merely approached—it is attained.

2. **A worst-case operating condition exists.**
   A continuous stress, temperature, error or vibration function on a bounded operating range reaches its largest value.

3. **Numerical optimization is meaningful.**
   The theorem justifies searching for an actual optimum, although it does not guarantee that a particular numerical algorithm will find it.

In several variables, the engineering form is:

> If $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and the feasible set $K \subset \mathbb{R}^n$ is nonempty and compact, then f attains a maximum and minimum on K.

In finite-dimensional engineering design, "compact" usually means that the feasible set is **closed and bounded**. This higher-dimensional version is one of the basic theoretical foundations of constrained optimization.

...ending

Main ideas to remember

- Continuity means

$$\lim_{x \to a} f(x) = f(a).$$

- Continuous functions preserve smooth input-output behaviour.
- A function must be one-to-one to possess an inverse.
- A continuous, strictly monotonic function on an interval has a continuous inverse.
- The Intermediate Value Theorem guarantees that continuous functions take every intermediate value.
- Weierstrass' theorem guarantees that a continuous function on a closed, bounded interval attains an absolute maximum and minimum.
- Weierstrass guarantees that extrema exist; derivative methods help locate them.

In-Class activity

If time allows…solve as many as possible from the following exercises (skip exercise 137)

SECTION 2.4 EXERCISES

For the following exercises, determine the point(s), if any, at which each function is discontinuous. Classify any discontinuity as jump, removable, infinite, or other.

131. $f(x) = \frac{1}{\sqrt{x}}$

132. $f(x) = \frac{2}{x^2+1}$

133. $f(x) = \frac{x}{x^2-x}$

134. $g(t) = t^{-1} + 1$

135. $f(x) = \frac{5}{e^x-2}$

136. $f(x) = \frac{|x-2|}{x-2}$

137. $H(x) = \tan 2x$

138. $f(t) = \frac{t+3}{t^2+5t+6}$